



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR **Symbiosis International (Deemed University)**

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

EXPERIMENT 3

AIM:

To implement JavaScript control structures and form validation by creating a grading system based on user-entered marks.

Software/Tools Required:

- Visual Studio Code (VS Code)
- Web Browser (Google Chrome/Microsoft Edge)
- HTML5
- JavaScript (ES6)

Theory:

JavaScript is a widely used scripting language that enables interactive and dynamic behavior on web pages. One of its key features is the use of control structures, such as if, else if, else, and switch, which allow programs to make decisions based on specified conditions. These control structures help in executing different blocks of code depending on the values provided by the user, making applications more intelligent and responsive.

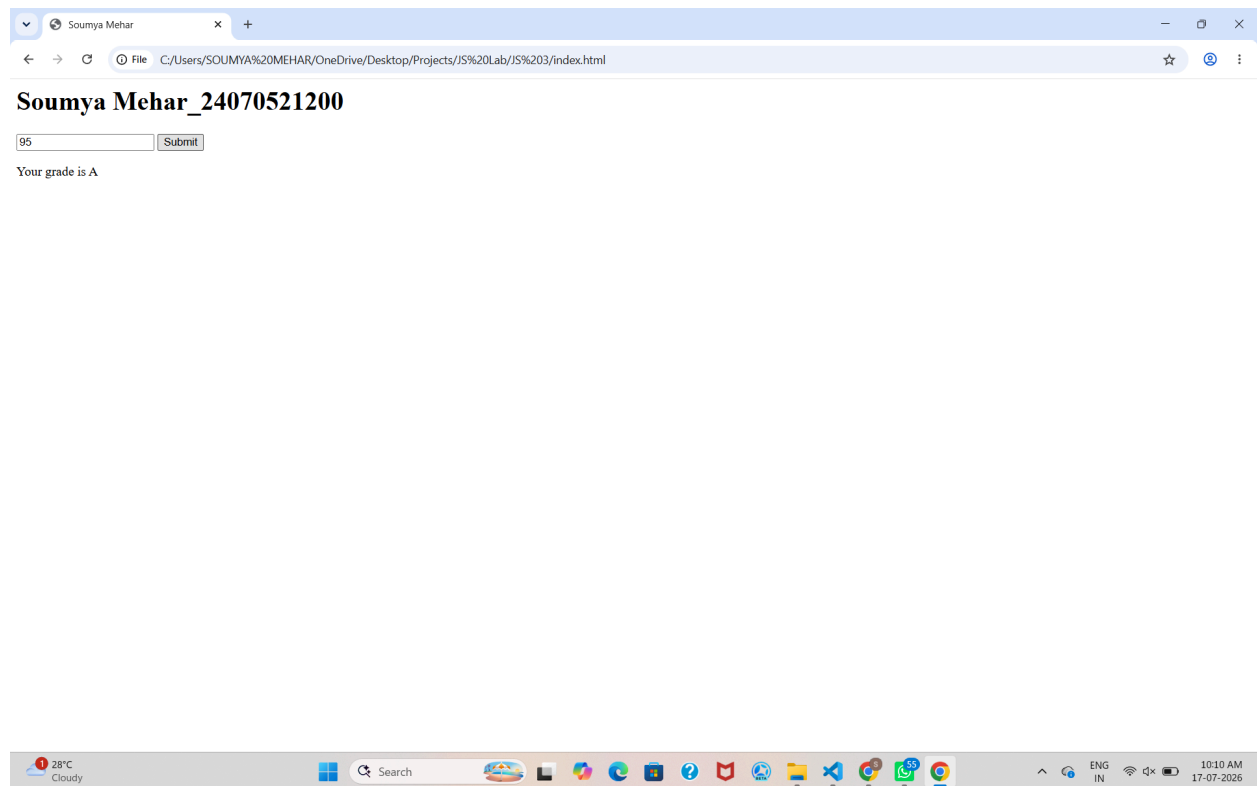
Another important concept in web development is form validation, which ensures that user input is accurate, complete, and within the expected range before it is processed. Validation helps prevent invalid or incomplete data from affecting the application's functionality. It checks conditions such as empty input fields, non-numeric values, and marks entered outside the valid range (0–100). This improves data integrity and enhances the overall user experience.

In this experiment, a grading system is developed using HTML and JavaScript. The user enters marks for different subjects through a form, and JavaScript validates the input before performing calculations. After successful validation, the program computes the total marks, percentage, grade, and pass/fail status. The grade is determined using conditional statements based on predefined percentage ranges, while the pass/fail status is decided according to the minimum passing criteria. This experiment demonstrates the



practical application of JavaScript control structures and form validation in building an interactive and reliable web application.

Output:



Result:

Successfully implemented JavaScript control structures and form validation to create a grading system that validates user input, calculates the total marks, percentage, grade, and pass/fail status accurately.



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CASE STUDY 3

AIM:

Scholarship Eligibility, A college awards scholarships based on marks. Below 50-Not Eligible, 50-69-Bronze, 70-84-Silver, 85-94-Gold, 95-100-Platinum.

PROGRAM CODE:

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8" />
5      <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6      <title>Soumya Eligibility Tester</title>
7      <style>
8          body {
9              font-family: Arial, sans-serif;
10             margin: 0;
11             padding: 40px 20px;
12             background: #f4f6f8;
13             display: flex;
14             justify-content: center;
15             align-items: center;
16             min-height: 100vh;
17         }
18
19         .card {
20             background: #fff;
21             padding: 24px;
22             border-radius: 12px;
23             box-shadow: 0 4px 12px rgba(0, 0, 0, 0.08);
24             width: 100%;
25             max-width: 420px;
26         }
```



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```
28 h1 {
29   font-size: 24px;
30   margin-bottom: 16px;
31   text-align: center;
32 }
33
34 label {
35   display: block;
36   margin-bottom: 8px;
37   font-weight: bold;
38 }
39
40 input {
41   width: 100%;
42   padding: 10px;
43   margin-bottom: 12px;
44   box-sizing: border-box;
45 }
46
47 button {
48   width: 100%;
49   padding: 10px;
50   border: none;
```



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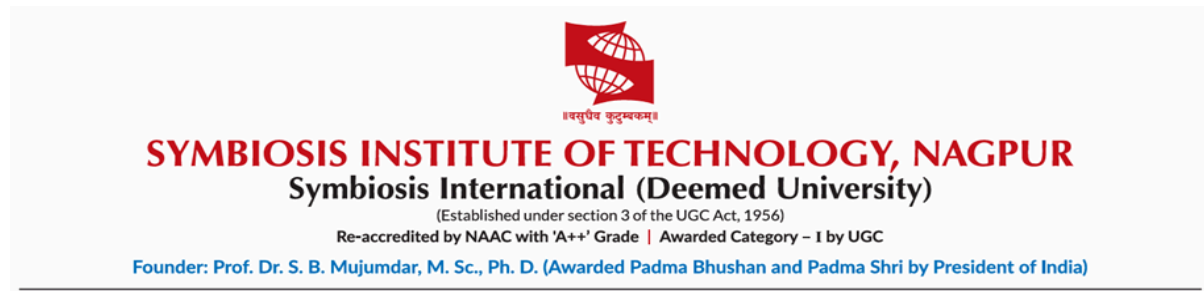
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```
67 <body>
68   <div class="card">
69     <h1>Soumya Eligibility Tester</h1>
70     <label for="marks">Enter marks</label>
71     <input type="number" id="marks" min="0" max="100" placeholder="e.g. 88" />
72     <button onclick="checkScholarship()">Check</button>
73     <div class="result" id="result">Result will appear here</div>
74   </div>
75
76   <script>
77     function checkScholarship() {
78       const marks = Number(document.getElementById('marks').value);
79       let message = 'Please enter a valid mark between 0 and 100.';
80
81       if (marks >= 0 && marks <= 100) {
82         if (marks < 50) {
83           message = 'Not Eligible';
84         } else if (marks <= 69) {
85           message = 'Bronze';
86         } else if (marks <= 84) {
87           message = 'Silver';
88         } else if (marks <= 94) {
89           message = 'Gold';
90         } else {
```



OUTPUT:

The screenshot shows a web browser window with a single tab titled 'Soumya Eligibility Tester'. The address bar shows the file path 'C:/Users/SOUMYA%20MEHAR/OneDrive/Desktop/Projects/JS%20Lab/Case%20Study%203/index.html'. The main content area displays a form titled 'Soumya Eligibility Tester'. The form has a label 'Enter marks' above a text input field containing the number '40'. Below the input field are two buttons: a blue 'Check' button and a light blue 'Not Eligible' button. The Windows taskbar is visible at the bottom, showing the system clock as 08:30 PM on 05-08-2026, and the weather as 29°C Mostly cloudy.

RESULT:

The Scholarship Eligibility was successfully implemented using JavaScript. The program validated the user input, assigned eligibility marks using if-else control structures, and displayed the correct Eligibility awards based on the entered marks. It effectively demonstrated the use of control structures and form validation in web development.